



An Application of Symmetry Methods to Study Gradient Flows of Error-Correcting Codes

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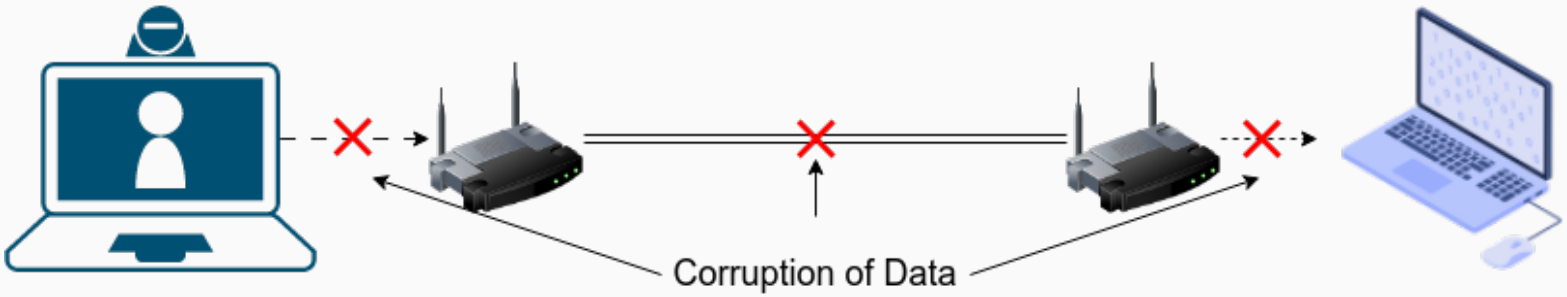


Raj Shekhar



Yingyao Zhou

Motivating ECCs



How do we reliably send the webcam data?

Naive: Improve our hardware - Difficult!

Shannon: Use bad hardware, but add redundancy to data - Easier!

Examples

- **English Language** : HELLO $\rightsquigarrow$ H_L_O $\rightarrow$ HELLO
- **Repetition** : 1 $\rightarrow$ 111 $\rightsquigarrow$ 101 $\rightarrow$ 111 $\rightarrow$ 1
- **Hamming(7,4)** : 1010 $\rightarrow$ 1010101 $\rightsquigarrow$ 1010111 $\rightarrow$ 1010101 $\rightarrow$ 1010

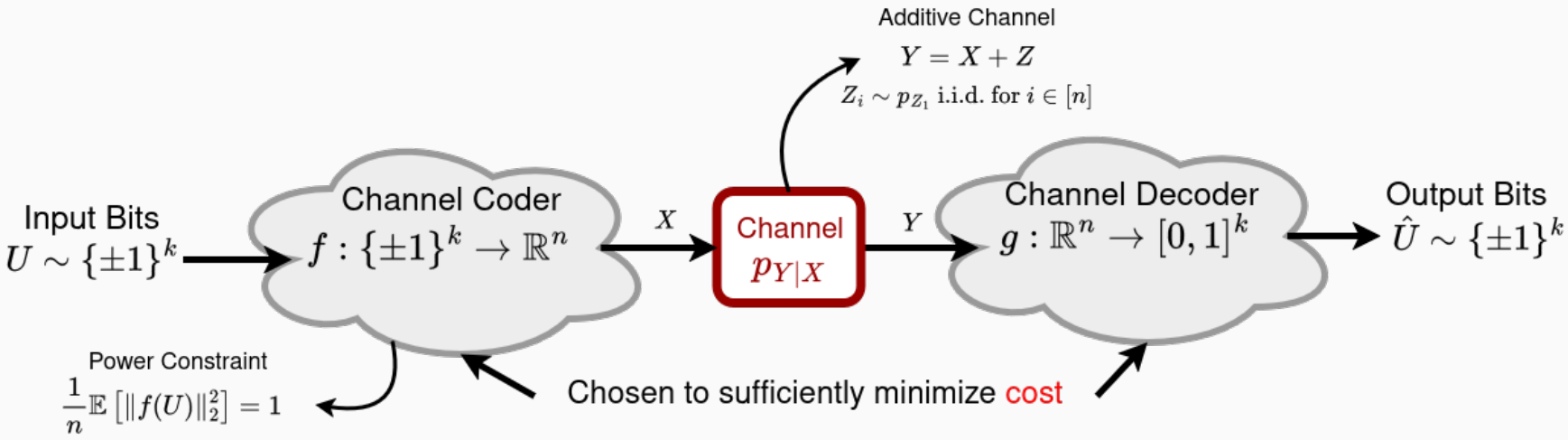
Shannon's Noisy Channel Theorem - Intuition from English

W__ther → Whether or Weather?

```
W__ther yo_ decode a m_s_age word-by-word, _ sin__e typo c_n
push _ou _ow__d the w_ong meaning and every subsequent gue_s
becomes mor_ fragi_e.  _f you in__e__ decod_
p_ra_raph-_y-paragr_ph, the s__rounding context _uppli_s
redunda_cy that lets you infer the intended wo_d_ even _hen
_e_eral _re _orrupted.  Both in Engl_sh and in
error-correcting codes, Shann_n's insigh_ is _h_t _i_h
e_o_gh struc_ured r_dun_ancy and a smart decod_r, re_i_ble
_ommuni_ation __ _os_ible even o_er a n_isy channel.
```

Shannon's Theorem (Informal): For any noisy channel, error rate $\epsilon > 0$, and rate R below channel-dependent capacity C, there exists a coding scheme of sufficiently large block length that allows communication at rate R with error $\leq \epsilon$.

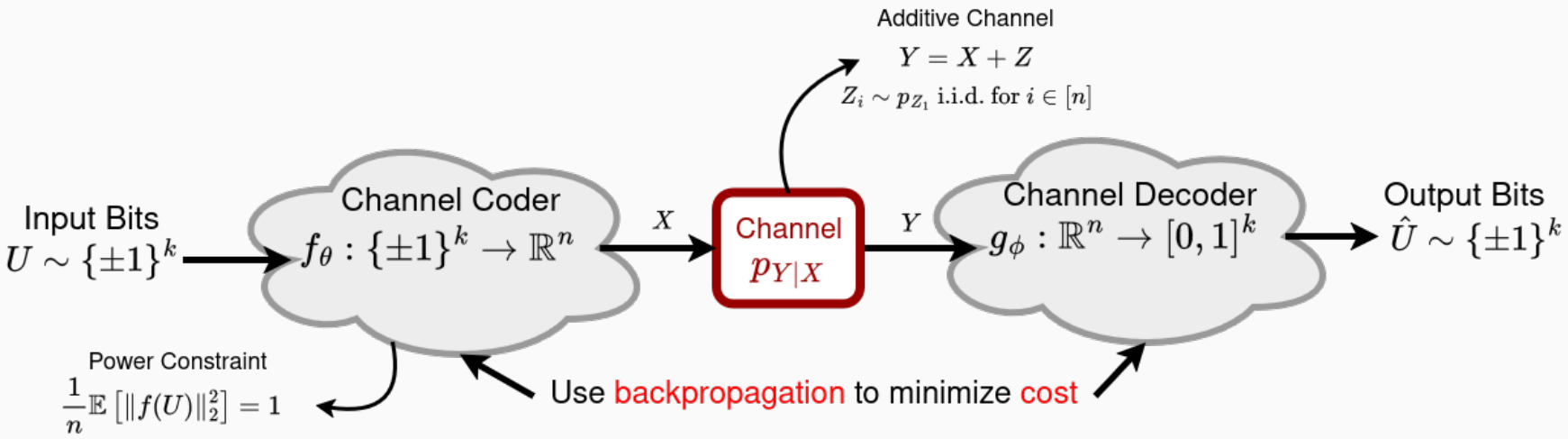
Formal Model of Communication



Examples of costs are

- *Bit Error Rate (BER)*: For $U \sim \{\pm 1\}^k$, the average number of mismatched bits between $\hat{U}$ and U
- *Binary Cross Entropy (BCE)*: The cross-entropy of $g_i(Y)$ wrt label U_i, averaged over $i \in [k]$

Deep-Learned Error Correcting Codes (DL-ECCs)



Examples

- TurboAE [6] "neuralizes" the class of *turbo codes* with convolutional neural networks.
- KO Codes [7] "neuralizes" the class of *Reed–Muller codes*.
- Error-Correction Code Transformer [2] "neuralizes" the class of *linear codes* with transformer-based architectures.

Why Explain DL-ECCs?

Traditional Codes: Human designed, guided by theory and engineering insight - Interpretable!
DL-ECCs: Black-boxes, learned from data, somewhat inspired by traditional codes - Not Interpretable!

How can we explain what these black boxes learned?

Explanations provide several benefits:

- *Scientific Insight*: What can we learn from the DL-ECC?
- *Debugging*: Which aspects are "features" and which are "bugs"?
- *Robustness*: Performance guarantees under perturbed conditions?
- *Compression*: Extract a simpler model that uses same "algorithm"?
- *Sandboxing*: Mathematically rigorous problem; can we use DL-ECCs as a sandbox for XAI?

One-Slide Primer on Boolean Analysis

For $S \subset [k]$, the *parity* on S is the function $\chi_S : \{\pm 1\}^k \to \{\pm 1\}$ defined as

$$\chi_S(u) = \prod_{j \in S} u_j \qquad \chi_S(u \odot v) = \chi_S(u)\chi_S(v),$$

where $\odot$ is the coordinate-wise product.

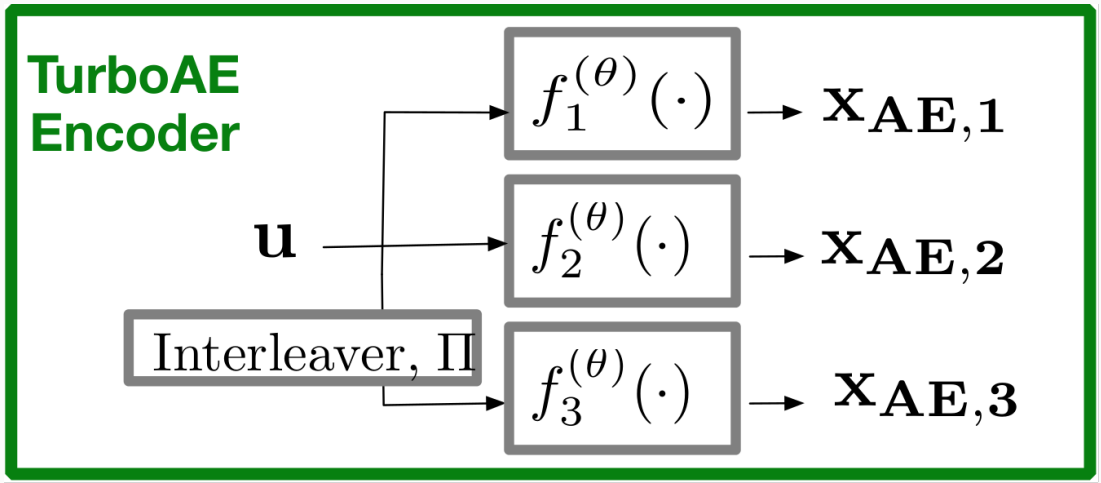
Any function $h : \{\pm 1\}^k \to \mathbb{R}$ can be expressed as a *linear combination* of parity functions:

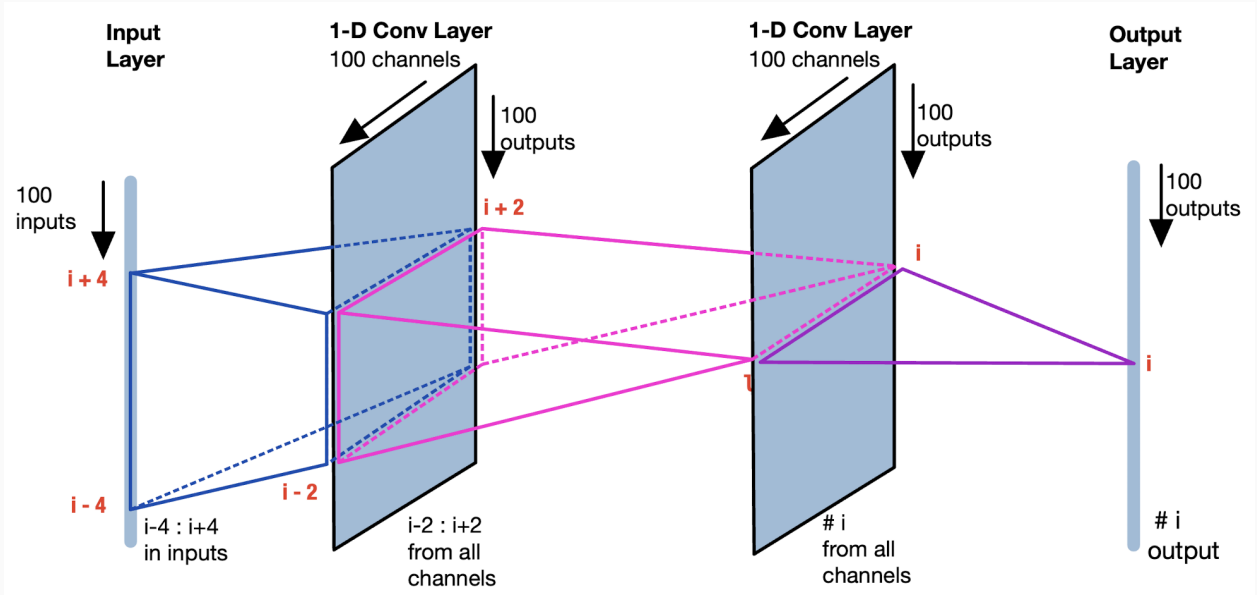
$$h = \sum_{S \subset [k]} \widehat{h}(S)\chi_S$$

called its Fourier decomposition. $|\widehat{h}(S)|$ measure how "close" h is to $\{\chi_S, -\chi_S\}$.

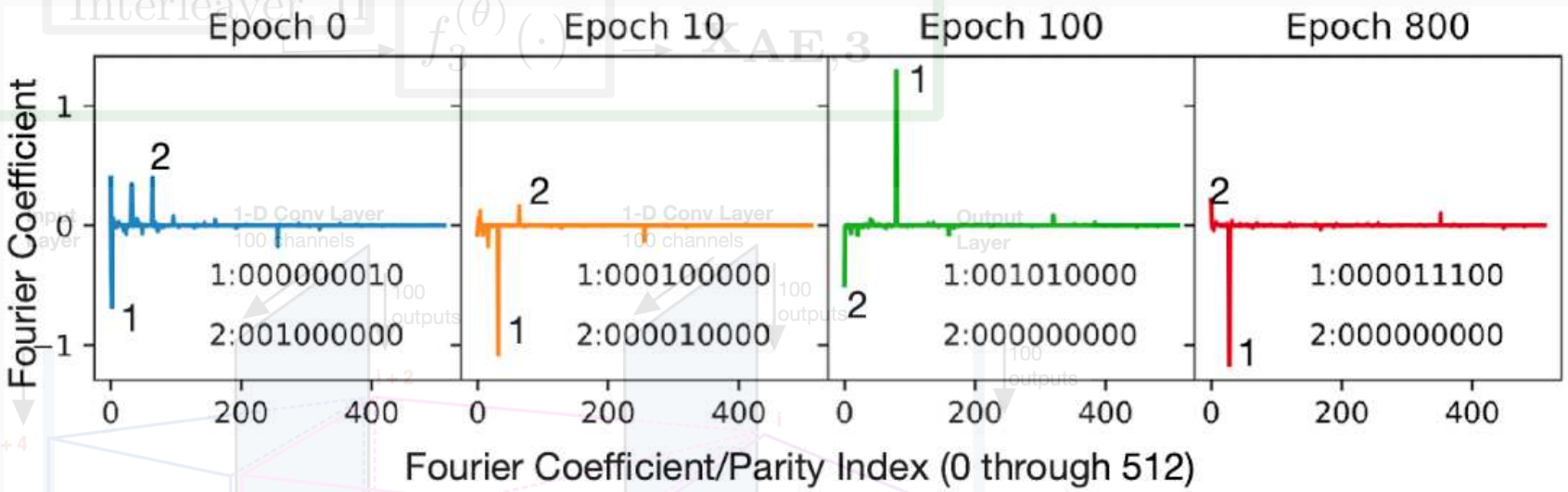
A *linear code* is a channel encoder $f : \{\pm 1\}^k \to \mathbb{R}^n$ whose components are parities.

Empirical Observations: TurboAE Encoder

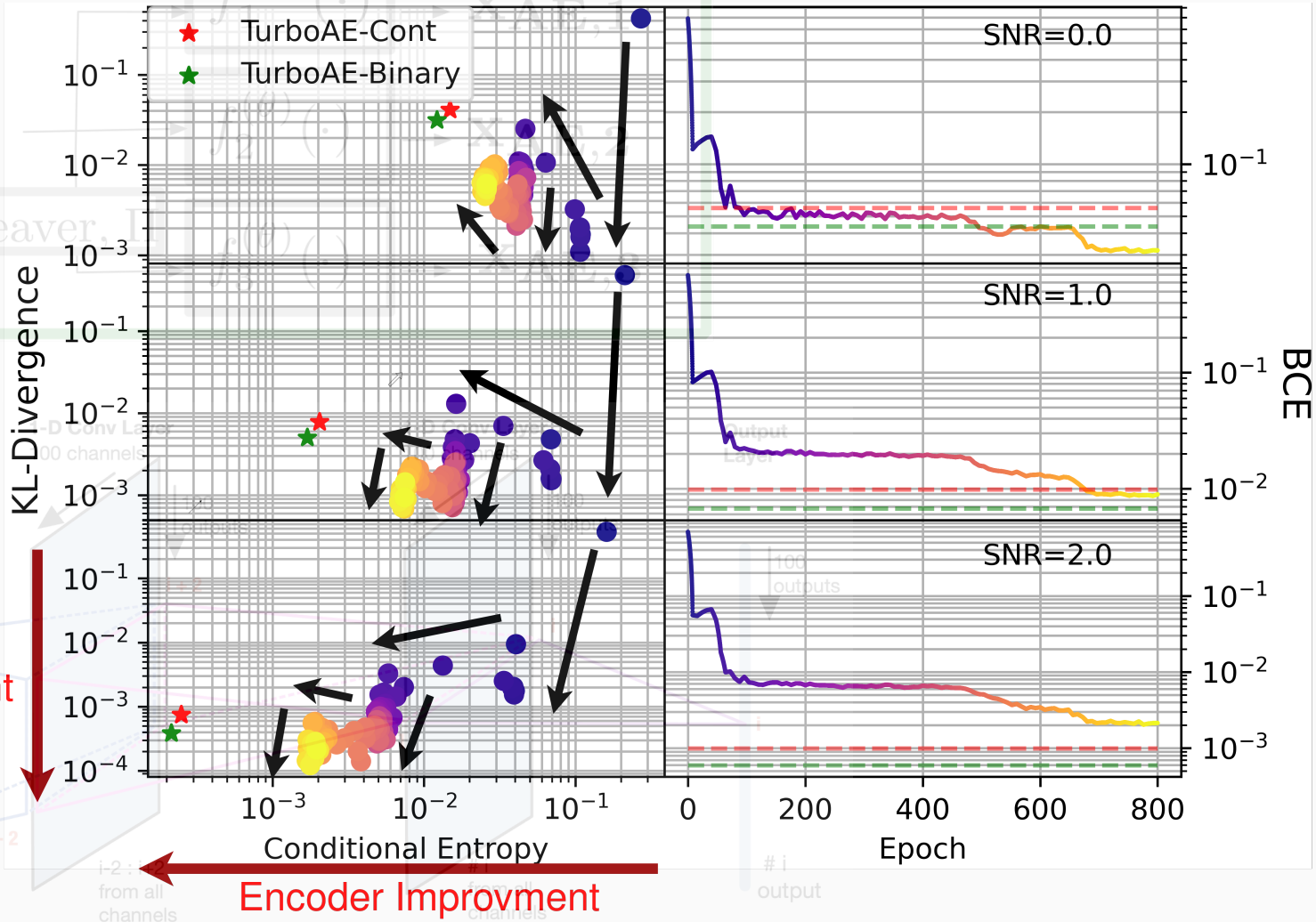




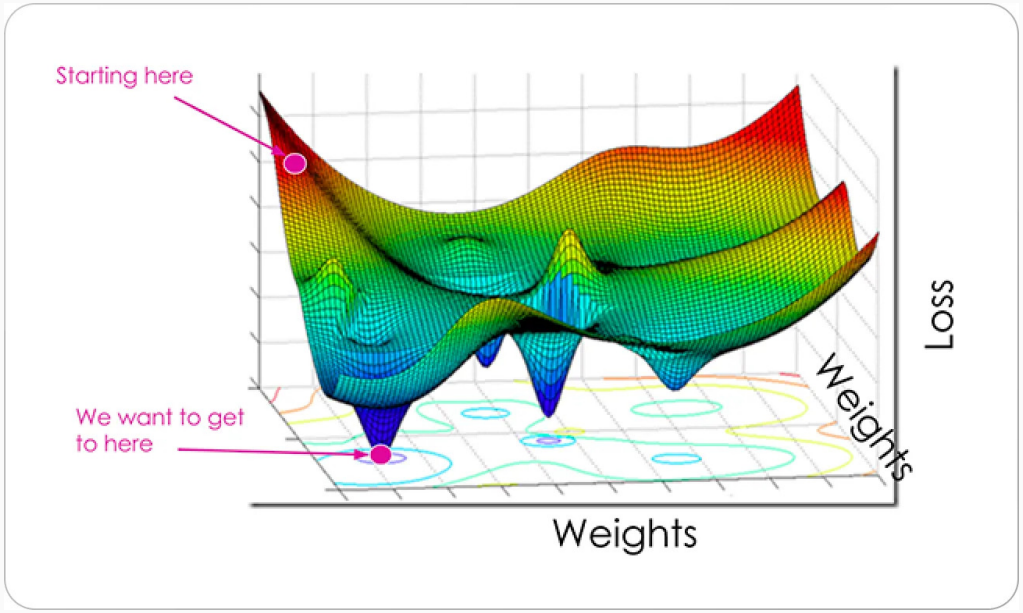
Empirical Observations: TurboAE Encoder



Empirical Observations: TurboAE Encoder



Loss Landscape

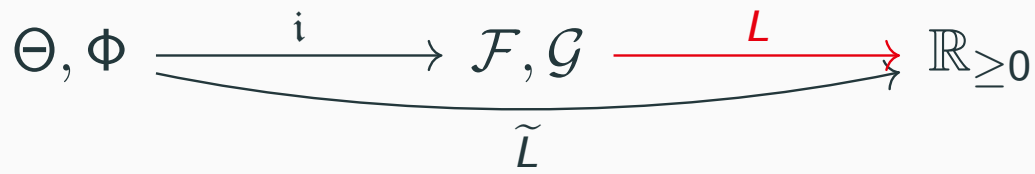


Source: Rosie Campbell

- How does the landscape govern what solutions DL-ECCs converge to?
- Can it explain some saddle-to-saddle behavior observed during training?
  - Is this behavior a consequence of using deep learning?

Decoupling Implementation and Function Space Landscapes

Key Idea: Decouple *problem induced* landscape from *neural network induced* landscape:



- Θ, Φ are the encoder, decoder network parameter spaces resp.
- $\mathcal{F}, \mathcal{G}$ are function spaces of encoders and decoders resp.
- $\mathfrak{i}$ is the *implementation* map: $\theta, \phi \mapsto f_\theta, g_\phi$
- L is loss on encoder-decoder pairs ($\widetilde{L} = L \circ \mathfrak{i}$ induced parameter loss)

Lemma (Informal)

Local minima (local maxima, critical points resp.) of L that are implemented by $\mathfrak{i}$ *pull back* to local minima (local maxima, critical points resp.) of $\widetilde{L}$ on parameters.

$\widetilde{L}$ inherits loss-landscape from L!

Removing the Decoder: Encoder-Space Loss

Focus: BCE loss

$$L(f,g) = -\frac{1}{k}\sum_{i=1}^{k}\mathbb{E}\left[U_i \log g_i(Y) + (1-U_i)\log(1-g_i(Y))\right].$$

Key Observation: For any encoder $f \in \mathcal{F}$,

$$L(f) := \min_{g\in\mathcal{G}} L(f,g) = L(f,p^{(f)}) = \frac{1}{k}\sum_{i=1}^{k}\mathbb{E}\left[h\left(p_i^{(f)}(Y)\right)\right],$$

where

- $p_i^{(f)}(y) = \mathbb{P}(U_i = 1 | Y = y)$ is the *MAP decoder* for f,
- $h(p) = -p\log p - (1-p)\log(1-p)$ is the binary entropy.

Studying L reduces to studying encoder-space loss equipped with MAP decoder!

Encoder Function Space: Details

Encoder function space: $\mathcal{F} = \{f : \{\pm 1\}^k \to \mathbb{R}^n\}$ with inner product

$$\langle f, g \rangle_{\mathcal{F}} \coloneqq \frac{1}{n} \mathbb{E}_{U \sim \{\pm 1\}^k} \left[\langle f(U), g(U) \rangle_{\mathbb{R}^n} \right].$$

Important Properties:

- $\mathcal{F}$ is a finite-dimensional *Hilbert space*, $\mathcal{F} \cong \mathbb{R}^{n2^k}$.
- Power constraint is expressed $\|f\|_{\mathcal{F}}^2 = 1$ (unit sphere $S(\mathcal{F})$).
- Parity components $\sqrt{n} e_i \chi_S$ for $i \in [n]$, $S \subset [k]$ form an orthonormal basis of $\mathcal{F}$.

Main Result

Hypothesis: Each linear code induces a critical point of the encoder-space loss L.

Theorem

Suppose $Z_i \overset{d}{=} -Z_i$ for all $i \in [n]$ and $L \in C^2(\mathcal{F})$. For any linear code $\chi : \{\pm 1\}^k \to \{\pm 1\}^n$, there exists $c \in \mathbb{R}^n$ s.t. $\|c\|_2^2 = n$ and $\lambda \in \mathbb{R}$ s.t.

$$\nabla L(c \odot \chi) = \lambda(c \odot \chi) \qquad \text{and} \qquad L(c \odot \chi) \leq L(\chi),$$

where $c \odot \chi := (c_1\chi_1, \ldots, c_n\chi_n)$.

$c \odot \chi$ is a *critical point* of L on the unit sphere $S(\mathcal{F})$ that is at least as good as χ.

Linear codes can attract gradient based algorithms!

Projected Gradient Flow

Goal: Prove theorem by gradient flow of L on unit sphere $S(\mathcal{F})$.

Projected gradient: for $f \in S(\mathcal{F})$,

$$\nabla_S L(f) = \nabla L(f) - \frac{\langle \nabla L(f), f \rangle_{\mathcal{F}}}{\|f\|_{\mathcal{F}}^2} f.$$

Projected gradient flow ODE:

$$\dot{f}_t = -\nabla_S L(f_t), \qquad f_t \in S(\mathcal{F}).$$

Critical point on $S(\mathcal{F}) \iff \nabla L(f) = \lambda f$ for $\lambda = \langle \nabla L(f), f \rangle_{\mathcal{F}}$.

Projected Flow Is Well-Behaved

Stays on the sphere: if $f_0 \in S(\mathcal{F})$ and $\dot{f}_t = -\nabla_S L(f_t)$, then

$$\frac{d}{dt}\|f_t\|_{\mathcal{F}}^2 = 2\langle f_t, \dot{f}_t\rangle_{\mathcal{F}} = -2\langle f_t, \nabla_S L(f_t)\rangle_{\mathcal{F}} = 0,$$

so $\|f_t\|_{\mathcal{F}} \equiv 1$.

Global existence/uniqueness: $S(\mathcal{F})$ is **compact** and $\nabla_S L \in C^1(S(\mathcal{F}))$. So initial $f_0 \in S(\mathcal{F})$ produces a unique global solution f_t for all $t \in \mathbb{R}$.

ω-limit points exist: compactness implies every trajectory has accumulation points. **ω-limit points are critical**.

Aside: Equivariant ODE Theory

Setup:

- ODE: $\dot{x} = F(x)$ for $F : \mathbb{R}^m \to \mathbb{R}^m$. Assume $F \in C^1$.
- Finite group Γ acting on $\mathbb{R}^m$ by isometries. Identify with finite subgroup of $O(m)$.
  - Example: $\Gamma = \{\pm 1\}^m$ acting on $\mathbb{R}^m$ by sign flips: for $\gamma \in \{\pm 1\}^m$ and $x \in \mathbb{R}^m$,

$$\gamma x = (\gamma_1 x_1, \gamma_2 x_2, \ldots, \gamma_m x_m).$$

  - Always possible to identify with subgroup of $O(m)$ by choice of inner-product and orthonormal basis.

Equivariance: The ODE is Γ-equivariant if for all $\gamma \in \Gamma$ and $x \in \mathbb{R}^m$,

$$F(\gamma x) = \gamma F(x).$$

F is Γ-equivariant $\iff$ for all $\gamma \in \Gamma$ and solutions $x(t)$, $\gamma x(t)$ is also a solution.

Aside: Equivariant ODE Theory

Setup:

- ODE: $\dot{x} = F(x)$ for $F : \mathbb{R}^m \to \mathbb{R}^m$. Assume $F \in C^1$.
- Compact Lie group Γ acting on $\mathbb{R}^m$ by isometries. Identify with compact Lie subgroup of $O(m)$.
  - Example: $\Gamma = \{\pm 1\}^m$ acting on $\mathbb{R}^m$ by sign flips: for $\gamma \in \{\pm 1\}^m$ and $x \in \mathbb{R}^m$,

$$\gamma x = (\gamma_1 x_1, \gamma_2 x_2, \ldots, \gamma_m x_m).$$

  - Always possible to identify with subgroup of $O(m)$ by choice of inner-product and orthonormal basis.

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F is Γ-equivariant $\iff$ for all $\gamma \in \Gamma$ and solutions $x(t)$, $\gamma x(t)$ is also a solution.

Aside: Equivariant ODE Theory

Isotropy subgroup: For $x \in \mathbb{R}^m$, the isotropy *subgroup* of x is

$$\Sigma_x = \{\gamma \in \Gamma : \gamma x = x\}.$$

Fixed point subspace: For a subgroup $\Sigma \leq \Gamma$, the fixed point *subspace* of Σ is

$$\mathrm{Fix}(\Sigma) = \{x \in \mathbb{R}^m : \gamma x = x, \forall \gamma \in \Sigma\}.$$

Theorem: If the ODE $\dot{x} = F(x)$ is Γ-equivariant,

1. $F(\mathrm{Fix}(\Sigma)) \subseteq \mathrm{Fix}(\Sigma)$ for any subgroup $\Sigma \leq \Gamma$
2. $\Sigma_{x(0)} = \Sigma_{x(t)}$ for any solution $x(t)$
3. $\Sigma_{x^*} \geq \Sigma_{x(0)}$ for any accumulation point x^*

Aside: Equivariant ODE Theory (Proofs)

Theorem

1. $F(\mathrm{Fix}(\Sigma)) \subseteq \mathrm{Fix}(\Sigma)$ for any subgroup $\Sigma \leq \Gamma$
2. $\Sigma_{x(0)} = \Sigma_{x(t)}$ for any solution $x(t)$
3. $\Sigma_{x^*} \geq \Sigma_{x(0)}$ for any accumulation point x^*

1. If $x \in \mathrm{Fix}(\Sigma)$, $\gamma \in \Sigma$, then $\gamma F(x) = F(\gamma x) = F(x)$.
2. If $\gamma \in \Sigma_{x(0)}$, then $\gamma x(0) = x(0)$; $x(t)$ and $\gamma x(t)$ solve same IVP, apply $\exists$ and !.
3. $x(t_n) \to x^*$, so for $\gamma \in \Sigma_{x(0)}$, $x^* \leftarrow x(t_n) = \gamma x(t_n) \to \gamma x^*$.

Remark

Viewpoint developed systematically by Golubitsky and collaborators to study bifurcations in symmetric dynamical systems; see [4, 1, 3].

Aside: Symmetry Breaking Example (S_3)

(Golubitsky–Stewart [4]): consider the ODE on $\mathbb{R}^3$ with parameter $\lambda \in \mathbb{R}$,

$$\begin{aligned}\dot{x} &= \lambda x - (x + y + z) + x^2,\\ \dot{y} &= \lambda y - (x + y + z) + y^2,\\ \dot{z} &= \lambda z - (x + y + z) + z^2.\end{aligned}$$

S_3-equivariance: $\Gamma = S_3$ acts by permuting coordinates.

Fully symmetric equilibrium: $\mathrm{Fix}(S_3) = \{(a, a, a) : a \in \mathbb{R}\}$. Choose $a = 0, 3 - \lambda$ gives equilibria on fixed subspace.

Symmetry breaking near 0: for $\lambda > 0$ small, equivariant bifurcation theory predicts branches of equilibria with isotropy conjugate to an S_2 subgroup: solutions lie in subspaces

$$\mathrm{Fix}(S_2) = \{(a, a, b) : a, b \in \mathbb{R}\}$$

and its three permutations. Solutions with fewer symmetries than ODE itself are called *symmetry-breaking* solutions.

Aside: Invariances $\Rightarrow$ Equivariances of Gradient Flow

Setup:

- $G : \mathbb{R}^m \to \mathbb{R}$ in C^2
- Finite group Γ identified with subgroup of $O(m)$.

Invariance: G is Γ-invariant if for all $\gamma \in \Gamma$ and $x \in \mathbb{R}^m$,

$$G(\gamma x) = G(x).$$

Theorem: If G is Γ-invariant, then

- Gradient flow ODE $\dot{x} = -\nabla G(x)$ is Γ-equivariant.
- Projected gradient flow ODE $\dot{x} = -\nabla_S G(x)$ is Γ-equivariant.

Aside: Invariances ⇒ Equivariances of Gradient Flow (Proofs)

Theorem

1. $\dot{x} = -\nabla G(x)$ is Γ-equivariant.
2. $\dot{x} = -\nabla_S G(x)$ is Γ-equivariant.

Proof: For $v \in \mathbb{R}^m$, $\gamma \in \Gamma$,

(1)

$$\begin{aligned}
&\langle \nabla G(\gamma x), v \rangle \\
&\quad = \frac{d}{dt} G(\gamma x + tv)\Big|_{t=0} \\
&\quad = \frac{d}{dt} G\big(\gamma(x + t\gamma^{-1} v)\big)\Big|_{t=0} \\
&\quad = \frac{d}{dt} G(x + t\gamma^{-1} v)\Big|_{t=0} \\
&\quad = \langle \nabla G(x), \gamma^{-1} v \rangle \\
&\quad = \langle \gamma \nabla G(x), v \rangle.
\end{aligned}$$

(2)

$$\begin{aligned}
&\nabla_S G(\gamma x) \\
&\quad = \nabla G(\gamma x) - \langle \nabla G(\gamma x), \gamma x \rangle \gamma x \\
&\quad = \gamma \nabla G(x) - \langle \gamma \nabla G(x), \gamma x \rangle \gamma x \\
&\quad = \gamma \left(\nabla G(x) - \langle \nabla G(x), x \rangle x \right) \\
&\quad = \gamma \nabla_S G(x).
\end{aligned}$$

General Technique

Symmetry method for gradient flows:

1. Identify a group Γ that leaves the loss invariant.
2. Conclude the (projected) gradient flow is Γ-equivariant.
3. Pick a structured point (here: a linear code χ) with a large isotropy subgroup.
4. Compute $\mathrm{Fix}(\Sigma_\chi)$: this gives an invariant low-dimensional subspace.
5. If ω-limit points exist, you get critical points also fixed by Σ_χ.

What Invariances Does L Have?

Two basic sign symmetries (assuming symmetric noise $Z_i \overset{d}{=} -Z_i$):

- **Flip codeword coordinates** (post-compose): for $x \in \{\pm 1\}^n$,

$$(x[f])(u) := x \odot f(u).$$

- **Flip input bits** (pre-compose): for $v \in \{\pm 1\}^k$,

$$(v[f])(u) := f(v \odot u).$$

Post-composition commutes with pre-composition:

$$(x[v[f]])(u) = x \odot f(v \odot u) = v[x[f]](u).$$

So identify group of symmetries as

$$\Gamma = \{\pm 1\}^n \times \{\pm 1\}^k.$$

Proof Idea: Why L Is Sign-Flip Invariant

Recall: we study the encoder-space loss $L(f) := L(f, p^{(f)})$ with $Y = f(U) + Z$, $U \sim \{\pm 1\}^k$, and $p_i^{(f)}(y) = \mathbb{P}(U_i = 1 \mid Y = y)$. Then

$$L(f) = \frac{1}{k} \sum_{i=1}^{k} \mathbb{E}\big[h\big(p_i^{(f)}(Y)\big)\big], \qquad h(p) = -p \log p - (1-p) \log(1-p).$$

(A) Post-compose $x \in \{\pm 1\}^n$:
Since

$$Z \overset{d}{=} x \odot Z,$$

mirroring codebook by x will *equivariantly* change

$$p_i^{(x[f])}(y) = p_i^{(f)}(x \odot y).$$

Expectation averages this out, so

$$L(x[f]) = L(f).$$

(B) Pre-compose $v \in \{\pm 1\}^k$:
If $v_i = -1$, then bit i assignments are flipped, so

$$p_i^{(v[f])}(y) = 1 - p_i^{(f)}(y).$$

h is *symmetric* about $1/2$, so

$$L(v[f]) = L(f).$$

Isotropy of a Linear Code χ

Let $\chi = (\chi_1, \ldots, \chi_n) : \{\pm 1\}^k \to \{\pm 1\}^n$ where each χ_i is a parity.

Defining identity for linear code:

$$\chi(v \odot u) = \chi(v) \odot \chi(u).$$

Claim:

$$\Sigma_\chi = \{A_v \in \Gamma : v \in \{\pm 1\}^k\}, \qquad A_v[f](u) = \chi(v) \odot f(v \odot u).$$

Flip inputs by v, and compensate by flipping codeword signs by $\chi(v)$:

$$(A_v[\chi])(u) = \chi(v) \odot \chi(v \odot u) = \chi(v)^2 \odot \chi(u) = \chi(u)$$

Converse easily checked too by setting $u = 1$.

Fixed Points of Σ_χ Are Exactly $c \odot \chi$

Claim

$$\mathrm{Fix}(\Sigma_\chi) = \{c \odot \chi : c \in \mathbb{R}^n\}.$$

Sketch: if f is fixed by every A_v, then

$$f(v) = \chi(v) \odot f(1) \;\Rightarrow\; f = c \odot \chi \text{ with } c = f(1).$$

Conversely, easy to check $c \odot \chi \in \mathrm{Fix}(\Sigma_\chi)$.

The projected gradient flow starting at χ stays in a **low-dimensional manifold**!

$$f_t \in \mathrm{Fix}(\Sigma_\chi) \cap S(\mathcal{F}) \;=\; \{c \odot \chi : \|c\|_2^2 = n\} \cong \mathbb{S}^{n-1}.$$

Conclude Existence of a Critical Point

Let f_* be any ω-limit point of the projected gradient flow starting at χ.

Standard gradient-flow fact: any accumulation point of projected gradient flow is a **stationary point**:

$$\nabla_S L(f_*) = 0 \quad \Longleftrightarrow \quad \nabla L(f_*) = \lambda f_*.$$

Since isotropy cannot decrease at limits, $f_* \in \mathrm{Fix}(\Sigma_\chi)$, hence

$$f_* = c_* \odot \chi \quad \text{with } \|c_*\|_2^2 = n.$$

This is exactly the theorem statement.

General Technique

Symmetry method for gradient flows:

1. Identify a group Γ that leaves the loss invariant.
2. Conclude the (projected) gradient flow is Γ-equivariant.
3. Pick a structured point (here: a linear code χ) with a large isotropy subgroup.
4. Compute $\mathrm{Fix}(\Sigma_\chi)$: this gives an invariant low-dimensional subspace.
5. Dynamics are restricted to that subspace. If you can deduce ω-limit points exist, you get critical points also fixed by Σ_χ.

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